

Week 1 · Student Notes

AI531C Digital Image Analysis · Tue 4 – Wed 5 Aug 2026 · Unit I · Lectures 1, 2, 3

Everything taught in this week, in one document. The per-lecture handouts are identical in content — use whichever you prefer.

What is in this bundle

- Lecture 1 — Course Orientation and the Digital Image-Analysis Pipeline
- Lecture 2 — Image Representation, Image Types, Spatial and Intensity Resolution
- Lecture 3 — Sampling, Quantisation, Aliasing and Quantisation Artefacts

This week at a glance

Sessions — Tue 4 – Wed 5 Aug 2026.

Tutorial — T1 (Tue 4 Aug).

Pre-class preparation — PCM-1.

Released after the last class of the week: Wed 5 Aug 2026, 18:00.

How to use this bundle — read the lecture section on the evening of the class while the argument is fresh, attempt the check-your-understanding questions before the tutorial, and leave the practice questions until you can do them unaided. No answer keys are printed: the reasoning is worked through in the tutorial.

Lecture 1 · Course Orientation and the Digital Image-Analysis Pipeline

AI531C Digital Image Analysis · Unit I · CO1 · M.Tech. Artificial Intelligence, Department of CSE, DTU

What this course builds: from classical image analysis to AI-based image analysis — and how your semester project grows with it

What you should already know

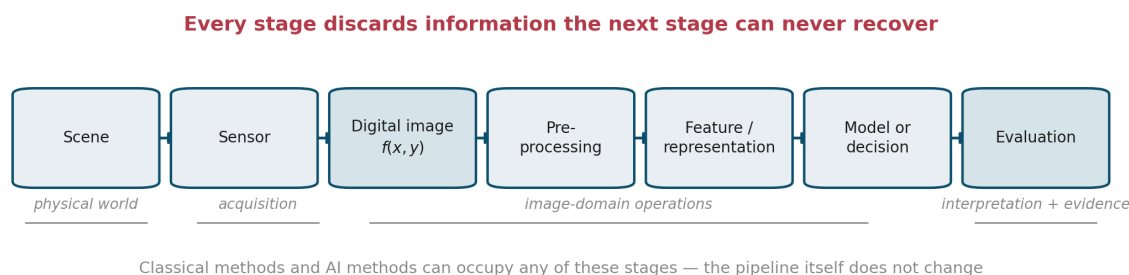
This is the first lecture of the course. No prior image-processing background is assumed.

Reading in the course notes: Comprehensive Notes L1-L2 (all 11 sections); Vol. I ch. 1-2.

Learning outcomes

After studying this lecture you should be able to:

Figure 1: The digital image-analysis pipeline



Course pipeline schematic. Source: course figure L1_F1.

How to read this figure

What to notice — Every project passes the same stages: scene → sensor → digital image → preprocessing → representation → decision → evaluation.

Why it happens — Each stage exists to remove one kind of uncertainty: geometry, noise, nuisance variation, semantics.

What it means — $f(x, y)$ is born at the sensor; everything after is transformation of f .

Why it matters — A failure downstream is usually caused upstream — debug in pipeline order.

The image-formation model

Problem: what does the number stored at pixel (x, y) represent? Assume a passive sensor viewing an illuminated scene, no colour, no motion.

$$f(x, y) = i(x, y) \cdot r(x, y)$$

What each symbol means

- $f(x, y)$ — recorded intensity at pixel (x, y) $i(x, y)$ — illumination falling on the scene point, $0 < i < \infty$ $r(x, y)$ — reflectance of the surface, $0 \leq r \leq 1$ $r = 0$: perfect absorber; $r = 1$: perfect reflector

How to read this

What it means — The camera never sees reflectance alone — every image entangles lighting with material. Most enhancement (histograms, gamma, low-light AI in Unit IV) is an attempt to undo unfavourable $i(x, y)$.

Worked example

Given — Grayscale image, $M \times N = 512 \times 512$, $b = 8$ bits/pixel; same image in RGB (3 channels).

Find — Storage in kilobytes for both.

Formula — $B = M \times N \times b / 8 \text{ bytes} (\times 3 \text{ for RGB})$

Solution

Substitution — $B = 512 \times 512 \times 8 / 8 = 512 \times 512$

Calculation — $B = 262\,144 \text{ bytes} = 262\,144 / 1024 \text{ KB}$ ANSWER: 256 KB grayscale; 768 KB RGB

What it means — Doubling M and N quadruples storage; adding channels multiplies it. A 4K video stream is this computation $\times$ frames — pipelines must budget data first.

The code behind the results

```
import numpy as np from skimage import data img = data.camera()  
# 2-D array print(img.shape, img.dtype)  
# (512, 512)  
uint8 print(img.min(), img.max())  
# 0 255 bytes_ = img.size * img.itemsize hist = np.bincount(img.ravel(), minlength=256)
```

Parameters used — dtype, shape order (rows, cols), channels. Full program:

L01_image_inspection.py

What it means — shape/dtype/range is the 10-second sanity check that prevents silent bugs.

Something to think about — The file were 16-bit? min/max $\approx 0\text{--}65535$, and every 255-based formula silently breaks.

Classical versus AI-based image analysis

Aspect	Classical (Units I-II)	AI-based (Units III-V)
Knowledge source	Designed by the engineer (models, kernels)	Learned from data (weights)
Data requirement	None to little	Curated datasets, labels or pairs
Interpretability	High — every step has a formula	Requires effort (feature maps, ablations)
Robustness	Brittle outside its assumptions	Robust within training distribution; fails silently outside it
Role in your project	Baseline + preprocessing + sanity check	Performance layer, compared against the baseline

Key takeaway: they are collaborators in one pipeline, not competitors — the course grades you on using both.

Where this is used — your project

This week you choose an application lane — low-light enhancement · denoising · deblurring · super-resolution · segmentation · fusion · RGB change detection

The same problem travels the whole semester — T8: classical baseline on it → T12: AI method on it → T14: integrated capstone on it. Today's action: write the one-sentence problem statement and the success criterion. That sentence is the seed of your capstone.

Checklist for a good lane — Data you can actually obtain; visible degradation or structure; a measurable success criterion; a classical method that plausibly works.

Core concepts

Common misconceptions — flagged on day one

“Enhancement and restoration are the same thing.” — No — restoration commits to a degradation model and can be objectively wrong; enhancement cannot.

“More megapixels means better analysis.” — Resolution beyond the task's requirement adds cost and noise, not information (L2–L3 make this quantitative).

“Deep learning makes classical methods obsolete.” — Every serious deployment keeps classical preprocessing, and every serious claim keeps a classical baseline.

“A high metric means a working system.” — Metrics summarise a distribution; failure cases live in its tail. This course marks failure analysis explicitly.

Summary

- An image is a sampled, quantised measurement $f(x, y)$ — illumination times reflectance. All course topics are stages of one pipeline; failures propagate downstream. Classical baselines are the evidence standard for every AI claim you will make.

Key formulae

$f(x, y) = i(x, y) \cdot r(x, y)$, $0 \leq r \leq 1$ $B = M \times N \times b / 8 \text{ bytes } (\times \text{ channels})$ Pipeline: scene $\rightarrow$ sensor $\rightarrow f(x, y) \rightarrow$ preprocess $\rightarrow$ represent $\rightarrow$ decide $\rightarrow$ evaluate

What comes next

Tutorial T1 — today 2:00–3:00 PM (studio) — Image representation and project orientation: load your own image; report shape, dtype, channels, histogram, bit depth; start the project-interest card.

Before Wednesday's double lecture (L2–L3) — Finish the PCM-1 artefact if incomplete. Bring one image whose storage you have computed by hand — we will spot-check three.

Logistics — Venue AB4-221. PCM-2 releases Friday 5 PM (sampling + colour detective, ~35 min). Next: L2 representation & resolution; L3 sampling & aliasing.

Check your understanding

Work these before the tutorial. They are the same questions used in class; the reasoning is discussed there, so attempt them first.

1. Diagnostic: audit your application

From your PCM-1 image: 1. What sensor produced it, and what did each pixel measure? 2. What is the most likely degradation? 3. What output does the application actually need — an image, a mask, a number, or a decision? 4. How would you measure success?

2. Which operation is it?

Classify each task — enhancement, restoration, segmentation, or analysis: A. Remove the stripe pattern a damaged sensor added to every frame. B. Make a dim endoscopy video watchable for the surgeon. C. Outline every building footprint in a satellite tile. D. Count how many buildings changed since last year.

Practice questions

Attempt these unaided, then compare with the discussion in your tutorial session.

Q1 (4 marks, understand): Distinguish image enhancement from image restoration, giving one example where the same operation serves each role.

Q2 (6 marks, apply): A camera produces 1920×1080 RGB frames at 10 bits/channel, 30 frames/s. Compute the raw data rate in MB/s and comment on its pipeline implications.

Q3 (10 marks, analyse): A factory inspection system worked in the lab but fails on the production line each evening. Using the pipeline model and $f = i \cdot r$, analyse the most likely cause and propose a two-step diagnostic.

Further reading

GW4 Ch.1 Secs.1.1,1.4 & Ch.2 Secs.2.2-2.4; Szeliski Ch.1-2; Elgendy Ch.1 Secs.1.3-1.7

Course notes: Comprehensive Notes L1-L2 (all 11 sections); Vol. I ch. 1-2.

Runnable code for every figure and number in this lecture: 06_CODE_AND_DEMOS (see the file names quoted beside each figure).

AI531C Digital Image Analysis · student handout · every figure and numerical result in this document is the output of course code you can run yourself.

Lecture 2 · Image Representation, Image Types, Spatial and Intensity Resolution

AI531C Digital Image Analysis · Unit I · CO1 · M.Tech. Artificial Intelligence, Department of CSE, DTU

Binary, grayscale and colour images as arrays — and what resolution, bit depth and dynamic range really buy you

What you should already know

This lecture continues from Lecture 1, “Course Orientation and the Digital Image-Analysis Pipeline”. Revise that material first — today's arguments assume it.

Reading in the course notes: Comprehensive Notes L2-L3; Vol. I ch. 2-3.

Learning outcomes

After studying this lecture you should be able to:

Intensity resolution and quantisation step

Problem: a sensor measures a continuous radiance range $[0, R]$. We store b bits per pixel. How finely can intensity be represented? Assume uniform quantisation.

$$L = 2^b \Delta = R / (L - 1)$$

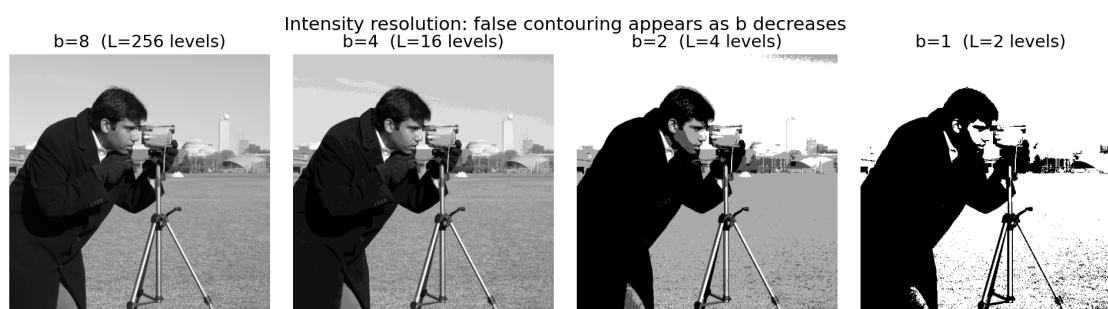
What each symbol means

- b — bits per pixel (bit depth) L — number of distinct intensity levels R — full measurable range (e.g. 255, or physical units) Δ — quantisation step: smallest representable difference

How to read this

What it means — Adding one bit doubles L and halves Δ . When Δ exceeds the local image contrast, neighbouring regions merge — visible as false contouring.

Figure 1: Reducing intensity resolution: false contouring



camera() quantised to $b = 8, 4, 2, 1$. Generated by L02_representation.py.

How to read this figure

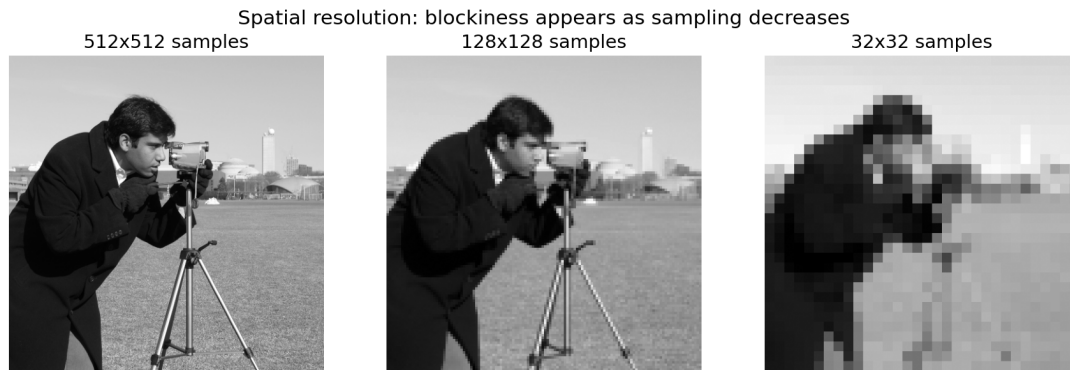
What to notice — Smooth sky regions break into visible bands as b drops below 5–6.

Why it happens — Step Δ grows beyond local gradients; many true values collapse to one level.

What it means — Contours are quantisation error made spatially coherent.

Why it matters — Diagnostic imaging and restoration targets need ≥ 10 bits; display needs only 8.

Figure 2: Reducing spatial resolution: blockiness



512², 128², 32² sampling of the same scene, displayed at equal size. Generated by L02_representation.py.

How to read this figure

What to notice — Fine structure (camera tripod, face) disappears first; large shapes survive.

Why it happens — Each sample now averages a larger scene patch; details smaller than the patch cannot be represented.

What it means — Spatial resolution bounds the finest recoverable feature — formalised tomorrow as the sampling theorem.

Why it matters — Match resolution to the smallest object the task must detect.

Worked example

Given — CT slice, 1024 × 1024 pixels, b = 10 bits, single channel.

Find — (a) Number of intensity levels L; (b) storage in MB.

Formula — $L = 2^b$; $B = M \times N \times b / 8$ bytes

Solution

Substitution — $L = 2^{10}$; $B = 1024 \times 1024 \times 10 / 8$

Calculation — $L = 1024$ levels; $B = 1\,310\,720$ bytes = $1\,310\,720 / 2^{20}$ MB ANSWER: $L = 1024$; $B = 1.25$ MB

What it means — A 500-slice study is ~625 MB — why PACS systems compress, and why bit depth is never 'free'.

Core concepts

Dynamic range: the third axis

Definition — Dynamic range = ratio of the largest to smallest distinguishable intensity, often quoted in dB: $DR = 20 \log_{10}(I_{\max} / I_{\min})$.

Distinct from bit depth — Bit depth is how finely we slice the range we can measure; dynamic range is how wide that range is. A sensor can have 12 bits and still clip highlights.

Consequences — Saturated pixels carry no recoverable information — no restoration method can invent them (revisited in Unit IV low-light and HDR-fusion contexts).

The code behind the results

```
q = (np.floor(img/255*L)
# quantise * (255/(L-1))).astype(np.uint8)
small = cv2.resize(img, (n, n), interpolation=cv2.INTER_AREA)
back = cv2.resize(small, (512, 512), interpolation=cv2.INTER_NEAREST)
```

Parameters used — L = levels; n = samples/side. Full program: L02_representation.py

What it means — Two independent dials: L controls banding, n controls blockiness.

Something to think about — Downsample with INTER_NEAREST instead? Aliasing — next hour's topic.

Image types at a glance

Type	Values / dtype	Typical use	Natural operations
Binary	{0, 1} / bool-uint8	Masks, documents, morphology	Erosion, dilation, CC labelling
Grayscale	0-255 / uint8 (or 10-16 bit)	Most analysis pipelines	Histograms, filtering, edges
Colour	3 × (0-255) / uint8, HWC	Natural scenes, remote sensing	Space transforms, per-channel ops
Float	[0, 1] / float32	Intermediate computation, ML	Anything without overflow

Key takeaway: convert deliberately (and document it) — silent uint8/float mixing is the classic day-one bug.

Where this is used — your project

This week, for your chosen lane, decide and record — input dtype and range convention (uint8 vs float [0,1]); working resolution — and what the smallest task-relevant structure is; bit depth of the source data, and whether your pipeline preserves it; channel layout (HWC vs CHW — PyTorch will want CHW in Unit III). These four lines go into your dataset card (PCM-11 will demand them).

Lane notes — Low-light: keep ≥ 10 -bit sources if possible; super-resolution: define HR/LR pair sizes now; change detection: equal size and depth across dates.

Summary

- An image is an array with a declared shape, dtype and range — declare them always. Spatial and intensity resolution fail differently: blockiness vs banding. Representation is a task-driven budget: buy only the bits the task needs.

Key formulae

$$L = 2^b \Delta = R / (L - 1) \quad B = M \times N \times b / 8 \quad (\times \text{ channels}) \quad DR = 20 \log_{10}(I_{\max} / I_{\min}) \text{ dB}$$

What comes next

Immediately next (L3, same room) — Sampling, quantisation and aliasing — we will reuse today's Δ and storage results quantitatively.

Carry-over exercise — From the L2 exercises: compute L and Δ for your own project imagery and note the smallest structure you must preserve — bring the two numbers to Tuesday's tutorial T2.

Logistics — Tutorial T2 (11 Aug): sampling, quantisation, colour and contrast studio. PCM-2 due Monday 8 PM.

Check your understanding

Work these before the tutorial. They are the same questions used in class; the reasoning is discussed there, so attempt them first.

1. Predict: which degradation hurts which task?

Two versions of the same document photo: A. 4× lower spatial resolution, full 8-bit depth. B. Full resolution, but 2-bit intensity depth. Predict which version is worse for: (1) reading small text; (2) separating paper from faint pencil marks. Explain using Δ and sample spacing.

Practice questions

Attempt these unaided, then compare with the discussion in your tutorial session.

Q1 (4 marks, remember/understand): Define spatial resolution, intensity resolution and dynamic range, and name the artefact produced by reducing each.

Q2 (6 marks, apply): A satellite pushbroom sensor delivers $12\,000 \times 12\,000$ pixels at 12 bits. Compute storage per scene in MB and per 14-scene orbit in GB.

Q3 (10 marks, analyse): A quality-control camera must detect 0.2 mm scratches on parts 100 mm wide while distinguishing 2% reflectance differences. Derive minimum resolution and bit depth, stating assumptions.

Further reading

GW4 Ch.2 Sec.2.4; Elgendy Ch.1 Secs.1.4-1.5; Forsyth-Ponce Part I

Course notes: Comprehensive Notes L2-L3; Vol. I ch. 2-3.

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Lecture 3 · Sampling, Quantisation, Aliasing and Quantisation Artefacts

AI531C Digital Image Analysis · Unit I · CO1 · M.Tech. Artificial Intelligence, Department of CSE, DTU

How many samples are enough — the sampling condition, what breaks when you violate it, and how to break it safely

What you should already know

This lecture continues from Lecture 2, “Image Representation, Image Types, Spatial and Intensity Resolution”. Revise that material first — today's arguments assume it.

Reading in the course notes: Comprehensive Notes L4; Vol. I ch. 4.

Learning outcomes

After studying this lecture you should be able to:

- state the two-dimensional sampling condition.
- predict when downsampling will alias, and what the artefact looks like.
- distinguish aliasing artefacts from quantisation artefacts.
- apply pre-filtering to downsample safely.
- solve sampling/storage trade-off numericals.

The sampling condition

Problem: a continuous scene contains spatial detail up to a maximum frequency $f_{\max}$ (cycles per mm). We sample it on a grid with rate f_s (samples per mm). When is perfect reconstruction possible? Assume band-limited signal and ideal sampling.

$$f_s > 2 \cdot f_{\max} \text{ (per axis)}$$

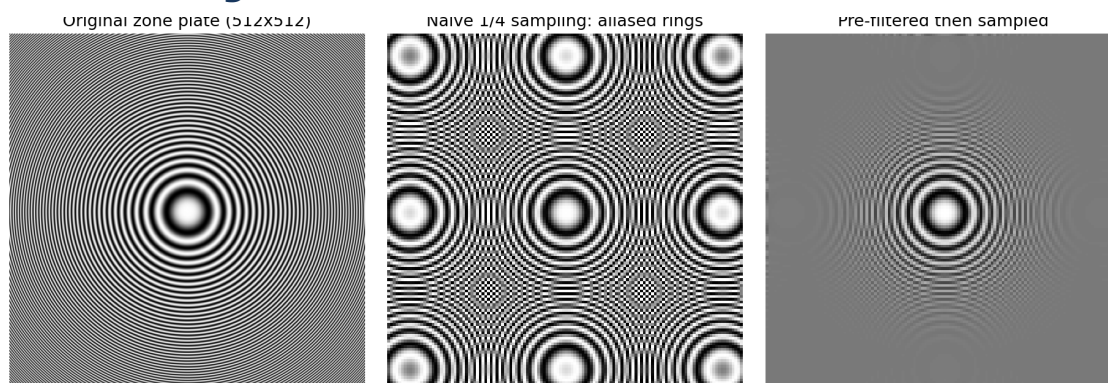
What each symbol means

- $f_{\max}$ — highest spatial frequency present (cycles/mm) f_s — sampling rate (samples/mm)
- $2 \cdot f_{\max}$ — the Nyquist rate: the minimum safe sampling rate Applies independently along x and y

How to read this

What it means — At least two samples must land on every finest cycle. Below the Nyquist rate, high frequencies do not disappear — they masquerade as false low frequencies. That impersonation is aliasing.

Figure 1: Aliasing made visible



Zone plate $512^2 \rightarrow$ naive $\frac{1}{4}$ sampling $\rightarrow$ pre-filtered sampling. Generated by L03_sampling_quantisation.py.

How to read this figure

What to notice — Naive sampling creates bold new rings that do not exist in the scene; pre-filtered sampling shows plain blur instead.

Why it happens — Ring frequencies above $f_s/2$ fold back to low frequencies (spectrum replicas overlap).

What it means — Aliasing is irreversible: once folded, false and true frequencies are indistinguishable.

Why it matters — Blur first, then sample (INTER_AREA).

Worked example

Given — Scanner output 4096×4096 at 8 bits. Finest document detail: 2 cycles/mm. Scanner sampled at 16 samples/mm.

Find — (a) Maximum safe integer downsampling factor s ; (b) storage before and after.

Formula — Need $f_s/s > 2 \cdot f_{\text{max}}$; $B = M \times N \times b / 8$

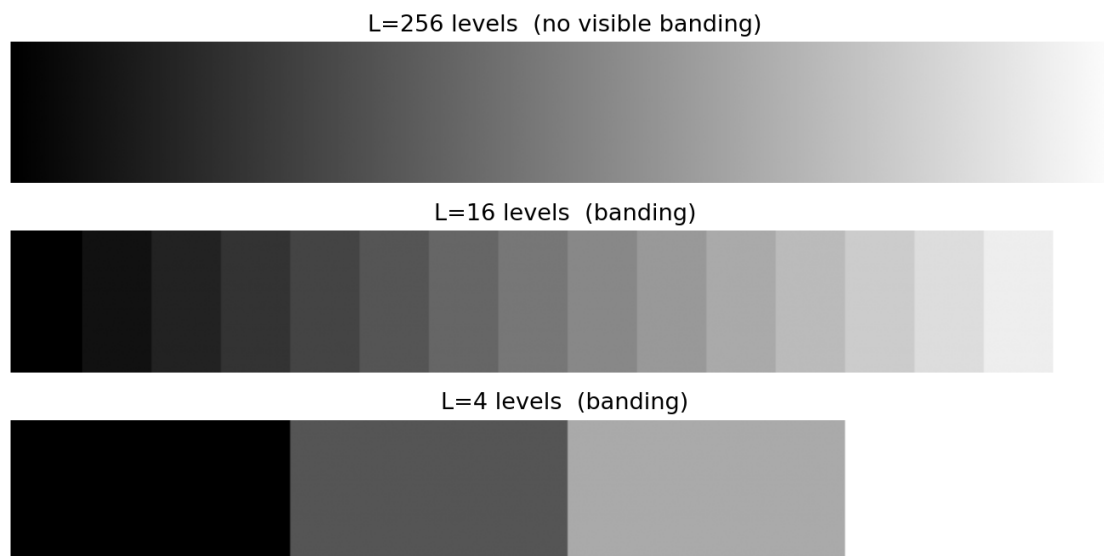
Solution

Substitution — $16/s > 4 \rightarrow s < 4 \rightarrow s = 3$ (integer)

Calculation — $B_{\text{before}} = 4096^2 = 16\,777\,216$ B = 16 MB; $B_{\text{after}} = (4096/3 \approx 1365)^2 \approx 1\,863\,225$ B ≈ 1.78 MB ANSWER: $s = 3$; 16 MB $\rightarrow \approx 1.8$ MB (9× saving)

What it means — The Nyquist condition converts a vague 'how small can I go' into an exact engineering bound — with an anti-alias filter before the resize.

Figure 2: Quantisation artefacts are a different animal



Smooth ramp at $L = 256, 16, 4$ levels. Generated by `L03_sampling_quantisation.py`.

How to read this figure

What to notice — Banding follows intensity contours, not spatial structure; edges stay sharp.

Why it happens — Quantisation is a pointwise operation on values; sampling is a spatial operation on positions.

What it means — Aliasing = spatial-frequency folding; banding = amplitude discretisation. Different causes, different cures.

Why it matters — Diagnose before treating: more bits fix banding; pre-filtering fixes aliasing; neither fixes the other.

The code behind the results

```
naive = zone[::4, ::4] # aliases blur = cv2.GaussianBlur(zone, (9, 9), 2.0)
safe = blur[::4, ::4] # clean # or, equivalently: safe2 = cv2.resize(zone, None, fx=.25,
fy=.25, interpolation=cv2.INTER_AREA)
```

Parameters used — factor 4; Gaussian $\sigma=2.0 \approx 0.5 \cdot s$. Full program: L03_sampling_quantisation.py

What it means — Same output size, radically different truthfulness.

Something to think about — σ too large $\rightarrow$ over-blur; too small $\rightarrow$ residual aliasing. The trade-off is continuous.

Aliasing vs quantisation artefacts — diagnosis table

Property	Aliasing	Quantisation banding
Cause	Sampling below Nyquist rate	Too few intensity levels
Appearance	False periodic patterns, moiré, jagged edges	Contour bands in smooth gradients
Where it strikes	Fine repetitive texture	Smooth low-gradient regions
Reversible?	No — frequencies already folded	No — but masked by dither/noise
Prevention	Pre-filter before sampling	More bits, or dithering

Key takeaway: identify which failure you have before choosing a remedy — they are not interchangeable.

Where this is used — your project

Every lane has a sampling question — Super-resolution: your LR images must be made by pre-filter + downsample, or you are training the model on aliasing; change detection: both dates must share one resolution — resample the finer one safely; denoising/low-light: patch extraction stride interacts with the finest structures you claim to preserve; segmentation: mask downsampling uses NEAREST — anti-aliasing a label map corrupts classes. Record your resize policy (filter, factor, library call) in the dataset card.

One-line audit — grep your pipeline for resize/slicing; for each hit, name the anti-alias step; no answer = latent bug.

Summary

- Two samples per finest cycle — or the scene lies to you. Aliasing folds frequencies (spatial); banding discretises amplitudes (intensity). Safe downsampling = deliberate blur + sample, never bare slicing.

Key formulae

$f_s > 2 f_{\max}$ (per axis) $M' = \lfloor M / s \rfloor$ $\Delta = R / (L - 1)$; error $\leq \Delta/2$ Pre-filter $\sigma \approx 0.5 \cdot s$ (rule of thumb)

What comes next

Tutorial T2 — Tuesday 11 Aug, 2–3 PM (studio) — Sampling, quantisation, colour and contrast studio: reproduce today's aliasing and banding panels on your own images; comparative output panel with interpretation is the artefact.

Pre-class (PCM-2, due Mon 8 PM) — Downsample and quantise one image at two levels; inspect RGB/HSV/Lab channels; three written observations (~35 min).

Logistics — Next lecture L4 (Tue 11 Aug, 12 PM): pixel neighbourhoods, connectivity, distance. L4 opens with a graded-style connectivity micro-problem — arrive having read the L4 note section 3.

Check your understanding

Work these before the tutorial. They are the same questions used in class; the reasoning is discussed there, so attempt them first.

1. Predict: the shrinking shirt

A student photographs a striped shirt (stripes ~ 3 px apart at full resolution) and creates a thumbnail with `img[::4, ::4]`. Predict: 1. What will the stripes look like in the thumbnail? 2. Where else in daily life have you seen this artefact? 3. Fix it with one line of code.

Practice questions

Attempt these unaided, then compare with the discussion in your tutorial session.

Q1 (4 marks, understand): State the 2-D sampling condition and explain, via spectrum replicas, why violating it is irreversible.

Q2 (6 marks, apply): An aerial sensor at 20 cm ground sampling distance must map ploughing furrows 0.5 m apart. Verify the Nyquist condition and give the safety margin.

Q3 (10 marks, analyse): A colleague downsamples $8\times$ with array slicing, sees stripes, and proposes median filtering the OUTPUT to remove them. Analyse why this fails and design the correct pipeline.

Further reading

GW4 Ch.2 Sec.2.4; Szeliski Ch.3 (multiresolution/sampling)

Course notes: Comprehensive Notes L4; Vol. I ch. 4.

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